



Welcome to ECE 222

Electrodynamics I

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- Get to Know the Instructor and TAs
- Course Overview
- Week 1 Content (review of vector calculus)

Why is it important to study Electromagnetics?

Electricity, magnetism, and electromagnetics really form the fundamentals of the electrical engineering field. Most of the concepts that we have seen, e.g. voltage, current, etc., have very beautiful EM explanations.

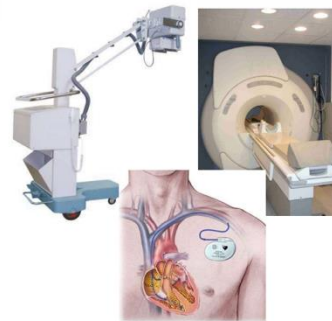
Very practical reason: Electromagnetic fields have a wide range of applications in our world today.



Power Generation



Telecommunications



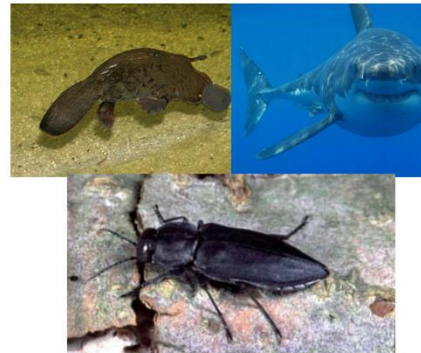
Healthcare



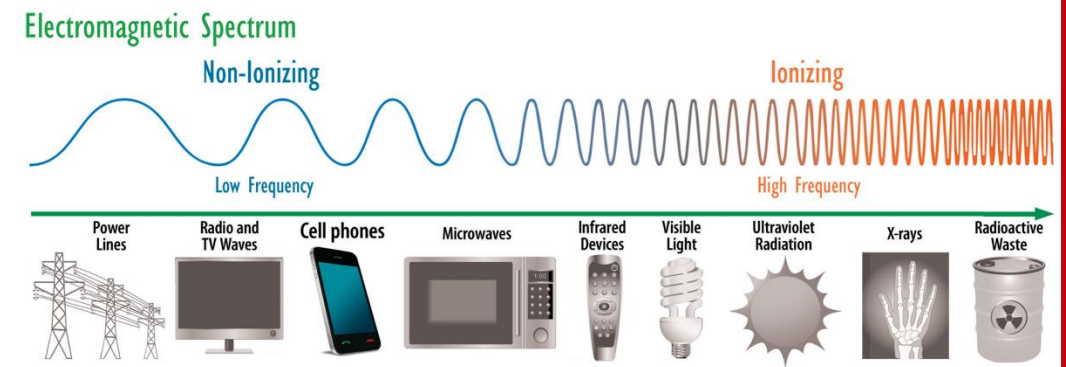
Aerospace & Defense



Radio astronomy



EM in Nature



Source: U.S. National Institute of Environmental Health Sciences

What Will We Learn in ECE 222?



Fundamentals of Electricity and magnetism and their relation to current and emerging technologies

- Vector calculus
- Electrostatics
- Magnetostatics
- Time varying electric and magnetic fields
- Electromagnetic waves
- Transmission lines

Course Information



Prerequisites

(PHYSICS 202, 208, or 248) and (MATH 234 or 376) or member of Engineering Guest Students.

The following math knowledge will be heavily used in ECE 222:

Calculus of functions of several variables; Calculus on parameterized curves, derivatives of functions of several variables, multiple integrals, vector calculus, three-dimensional coordinate systems (cartesian, cylindrical, and spherical), divergence, curl, and gradient operators, integration of vector fields of multiple variables along lines and surfaces.

We will have a quick review of some of these vector calculus topics in the first week of the course but this will not be enough, please revisit the concepts in Math 234 as needed.

Course Information



Nature of the Topics Covered in ECE 222

- Abstract and difficult to visualize.
- Difficult to understand the first time you encounter them — **this is absolutely normal.**
- Grasping these concepts requires time, practice, and effort.

How to Succeed in ECE 222?

- Review Math 234 topics as needed, particularly vector calculus, and make sure you are comfortable with the required mathematics.
- Attend all lectures and practice sessions.
- Ask questions whenever something is unclear. Take advantage of office hours and other course resources whenever you need help.
- Plan to spend 8+ hours per week outside of class studying and practicing ECE 222 material.
- Complete the assigned homework on time and independently.
- If a concept is still unclear, go beyond the assigned homework and solve additional practice problems.

Class

(Refer to Syllabus for more details)

- **Class will be conducted in-person**

Room: Engineering Hall 1227

Time: MWF 4:00 AM – 5:15 PM

- **Textbook:** Fundamentals of Applied Electromagnetics. 7th Edition.
By: Fawwaz Ulaby, Umberto Ravaioli. (eTextbook: <https://www.pearson.com/en-us/subject-catalog/p/fundamentals-of-applied-electromagnetics/P200000003174/9780135200445>)

Key topics are covered in

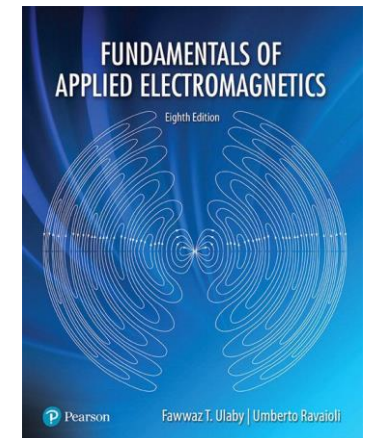
Chapter 3: Vector Analysis

Chapter 4: Electrostatics

Chapter 5: Magnetostatics

Chapter 6: Maxwell's Equations for Time-Varying Fields

A little bit of Chapter 7 (Electromagnetic waves) and Chapter 2 (Transmission lines)





Communication and Office Hours

(Refer to Syllabus for more details)

- We will use **Office hour** and **Piazza** as a main mode for contact/questions and responses
 - Link on the navigation bar on the Canvas page
 - Opportunity to interact with fellow students
 - ***Helping each other is the best way to learn!***
 - Faster response

- We have office hours **everyday**, so feel free to reach out for help!

Haihan's office hours

Monday, 5:15PM - 6:00PM	Engineering Hall 3441
Wednesday, 5:15PM - 6:00PM	Engineering Hall 3441
Friday, 5:15 PM - 6:00 PM	Engineering Hall 3441

TA's office hours

Tuesday, 1:00 to 2:00 PM	EH 2421	Kaimeng
Thursday, 1:00 to 2:00 PM	EH 2421	Kaimeng
Friday, 12:00 to 1:00 PM	EH 2421	Kaimeng
Wed, 12:00 - 1:00 PM	EH1422	Shwetambari

- Only use email if absolutely necessary. Include [ECE 222] in the subject line.



Assignments and Exams

(Refer to Syllabus for more details)

- **Homework (~11 sets, 10 pts each)**
 - Homework assignments are due on the due date indicated on the specific assignments.
Late homework assignments will not be accepted.
Individual extensions will not be granted under normal circumstances.
 - Homework solutions should be neatly written, with all steps clearly shown and final answers boxed for easy identification
 - Homework must be submitted electronically via Canvas as a single PDF document.
 - HW are graded on a completion basis. The lowest grade will be dropped in the final grade calculation.
- **In-class Example Problems (~11 sets, 10 pts each)**
 - On Wednesdays/Mondays, we will do in-class example problems in small groups and all together.
 - These must be electronically uploaded to Canvas by **11:59 pm on the following day (Thursdays/Tuesdays)**. **Late In-Class Problem submission will not be accepted.**
 - They are graded on a completion basis. The lowest grade will be dropped in the final grade calculation.



Assignments and Exams

(Refer to Syllabus for more details)

- **Three Exams (100 pts each, 300 pts in total)**

Exam 1	100 pts	Tentative time: Friday, October 02, in class
Exam 2	100 pts	Tentative time: Friday, November 06, in class
Final Exam	100 pts	Friday, December 11, 5:05-7:05 pm, Location: TBD

- **Grading Scale (No Curve)**

In-class Problems	100 pts
Homework	100 pts
Midterm Exam 1	100 pts
Midterm Exam 2	100 pts
Final Exam	100 pts
Total Points	500 pts

>90% A	68-75 C
86-89 AB	60-67 D
80-85 B	<60 F
76-79 BC	

Learning Resources



- Office hours
- Textbook
- Supplemental textbook material: <https://em8e.eecs.umich.edu/>
- Classmates/study groups
- Google
- YouTube
- AI tools (use with caution)
- Lectures by other professors/schools

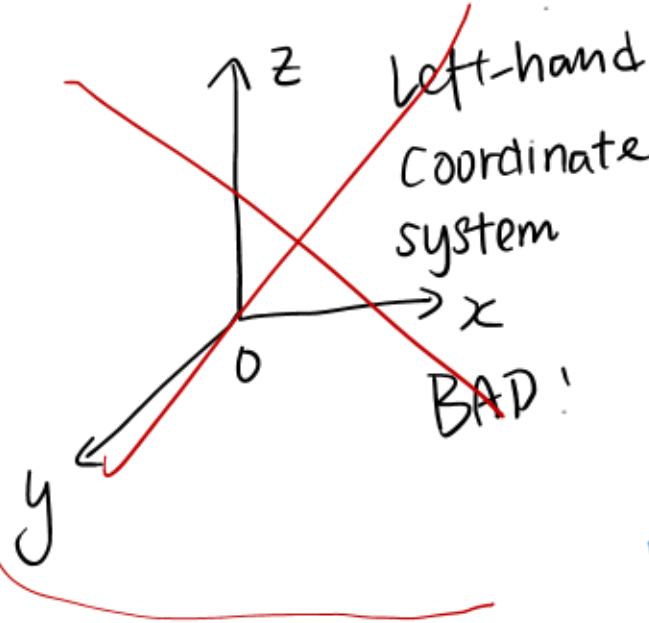
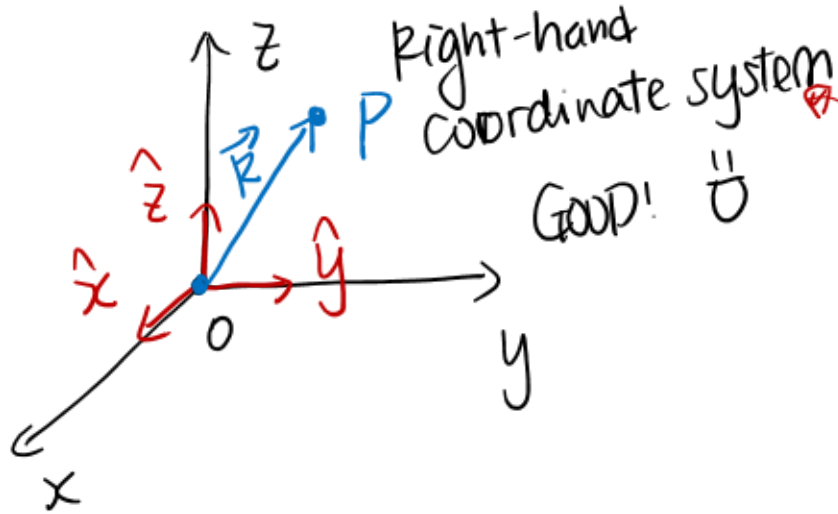


Hope you will enjoy our ECE 222!

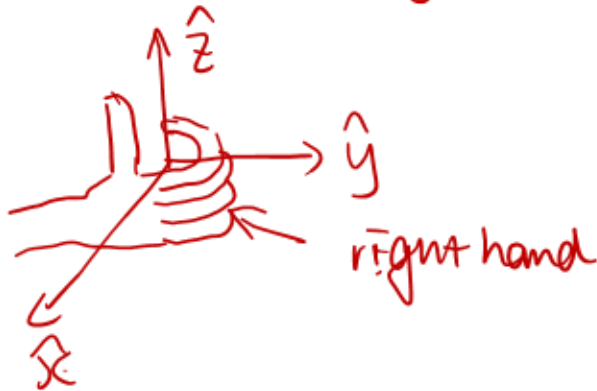
Review of Coordinate systems



Cartesian coordinate systems



unit vector $\hat{x}$ $\hat{y}$ $\hat{z}$, length 1
point at certain direction
no unit.



Point $P(x, y, z)$

e.g. $P(2m, 2m, 2m)$

Define the radius vector
as the position vector from
the origin O to P

$\vec{R} = \vec{OP} = \hat{x}x + \hat{y}y + \hat{z}z$
length of $\vec{R}$: $|\vec{R}| = \sqrt{x^2 + y^2 + z^2}$ scalar
cross-product convention:

$$\hat{x} \times \hat{y} = \hat{z}$$

$$\hat{y} \times \hat{z} = \hat{x}$$

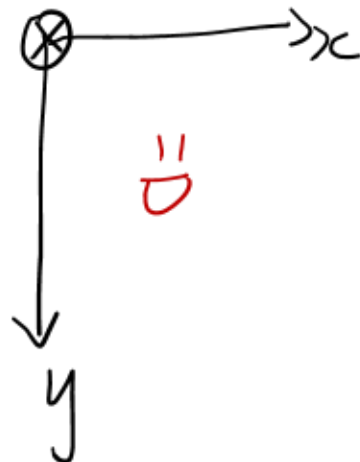
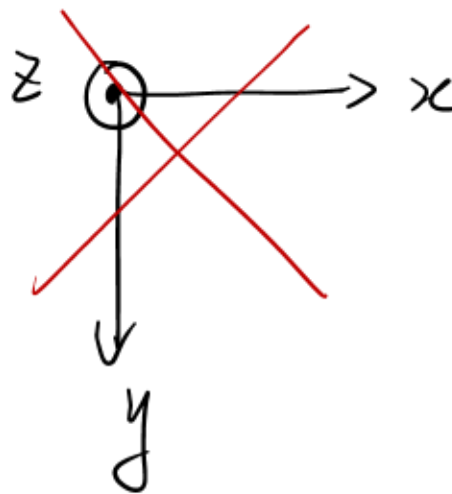
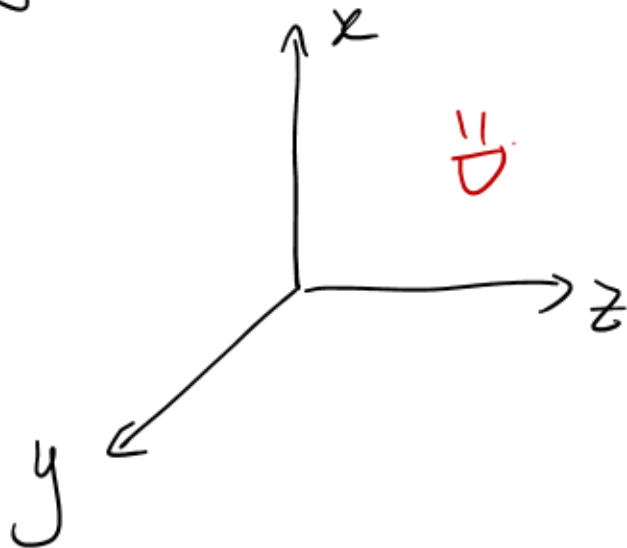
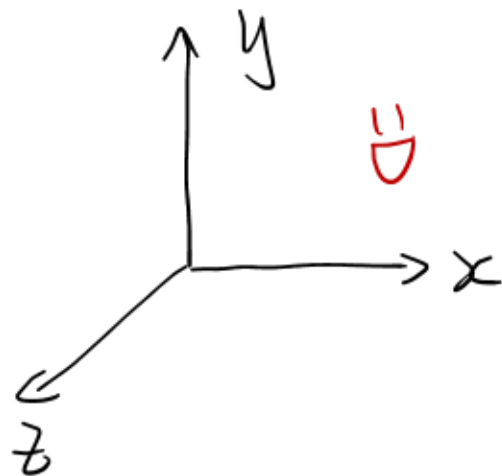
$$\hat{z} \times \hat{x} = \hat{y}$$



opposite
direction
gets a negative
sign!

Which one is good?

Which one is bad?

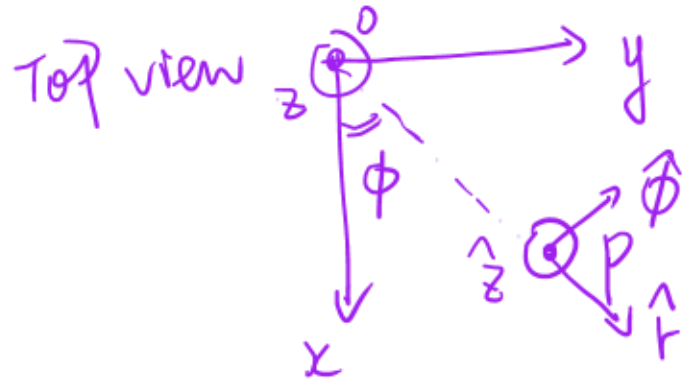
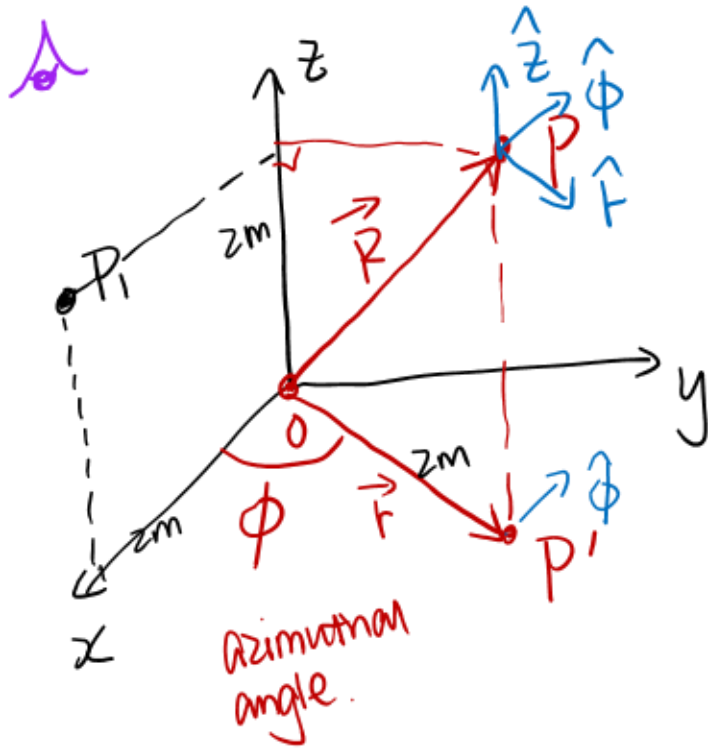


pointing out of the screen
 $\odot$

into the screen
 $\otimes$



Cylindrical Coordinate System



Radius Vector

$$\vec{R} = \vec{OP} = \hat{x}x + \hat{y}y + \hat{z}z$$

$$\vec{r} = \hat{r}r$$

$$= \hat{r}r + \hat{z}z$$

ϕ is made in $\hat{r}$
unit vector $\hat{r}$ $\hat{\phi}$ $\hat{z}$

Cross-product convention

$$\hat{r} = \hat{x} \cos \phi + \hat{y} \sin \phi$$

then $\hat{\phi}$?

$$\begin{aligned} \hat{\phi} &= \hat{z} \times \hat{r} \\ &= \hat{z} \times (\hat{x} \cos \phi + \hat{y} \sin \phi) \\ &= \hat{y} \cos \phi - \hat{x} \sin \phi \end{aligned}$$

$$\hat{r} \times \hat{\phi} = \hat{z}$$

$$\hat{\phi} \times \hat{z} = \hat{r}$$

$$\hat{z} \times \hat{r} = \hat{\phi}$$



$$P(r, \phi, z)$$

$$\text{e.g. } P(2m, 60^\circ, 2m)$$

$$P_2(2m, 0, 2m)$$

$$0 \leq r < \infty$$

$$0 \leq \phi < 2\pi$$

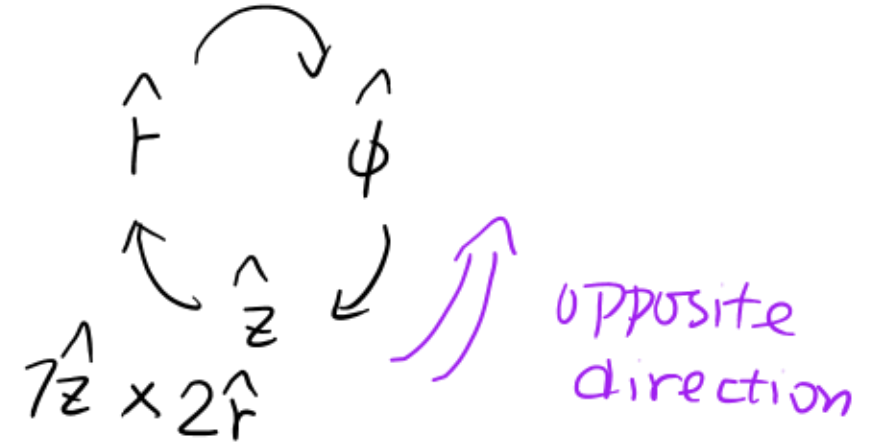
$$-\infty < z < \infty$$

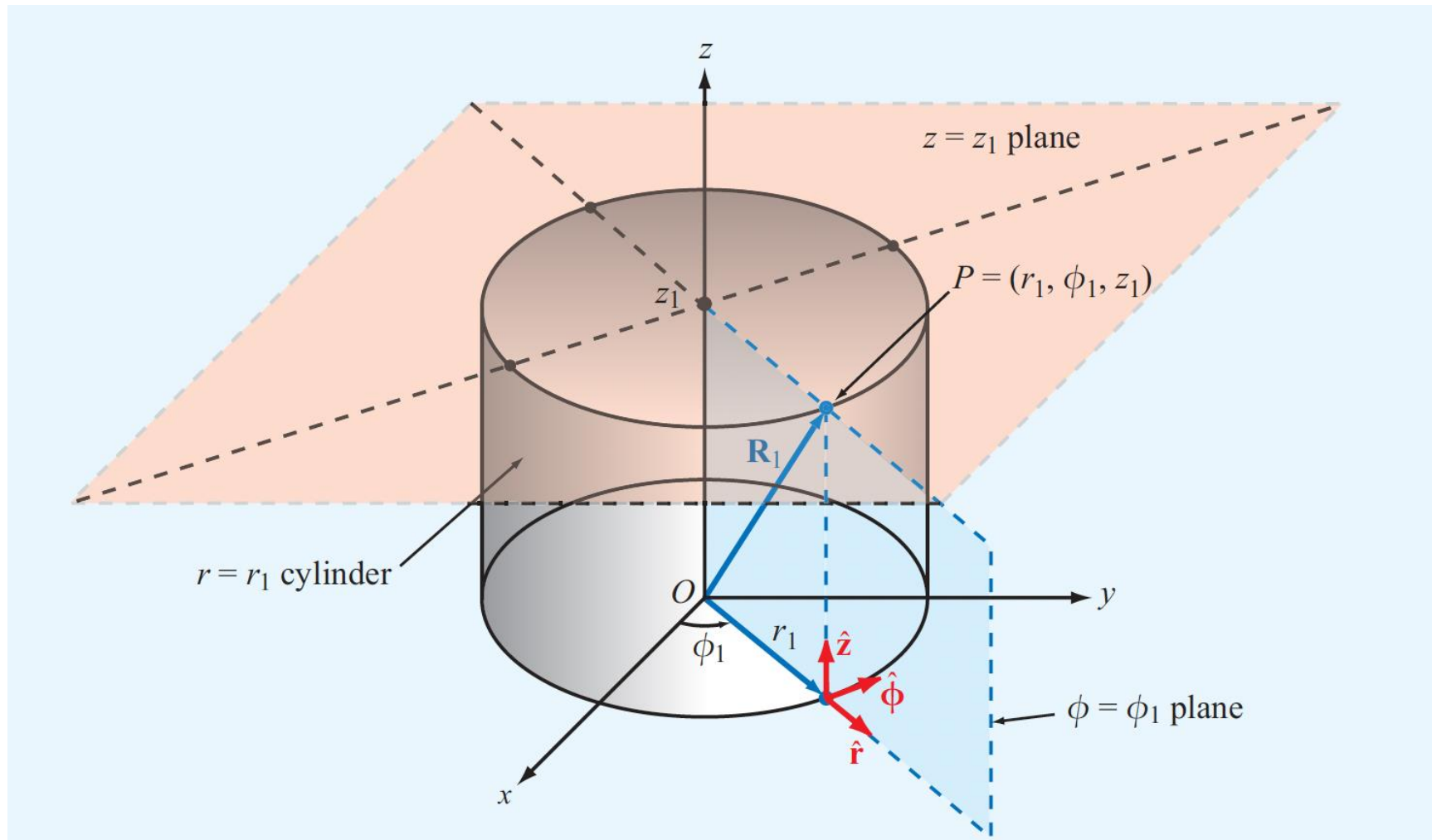
Example =

$$(5\hat{r} + 7\hat{z}) \times (10\hat{\phi} + 2\hat{r})$$

$$= 5\hat{r} \times 10\hat{\phi} + \cancel{5\hat{r} \times 2\hat{r}} + \underline{7\hat{z} \times 10\hat{\phi}} + 7\hat{z} \times 2\hat{r}$$

$$= 50\hat{z} - 70\hat{r} + 14\hat{\phi}$$



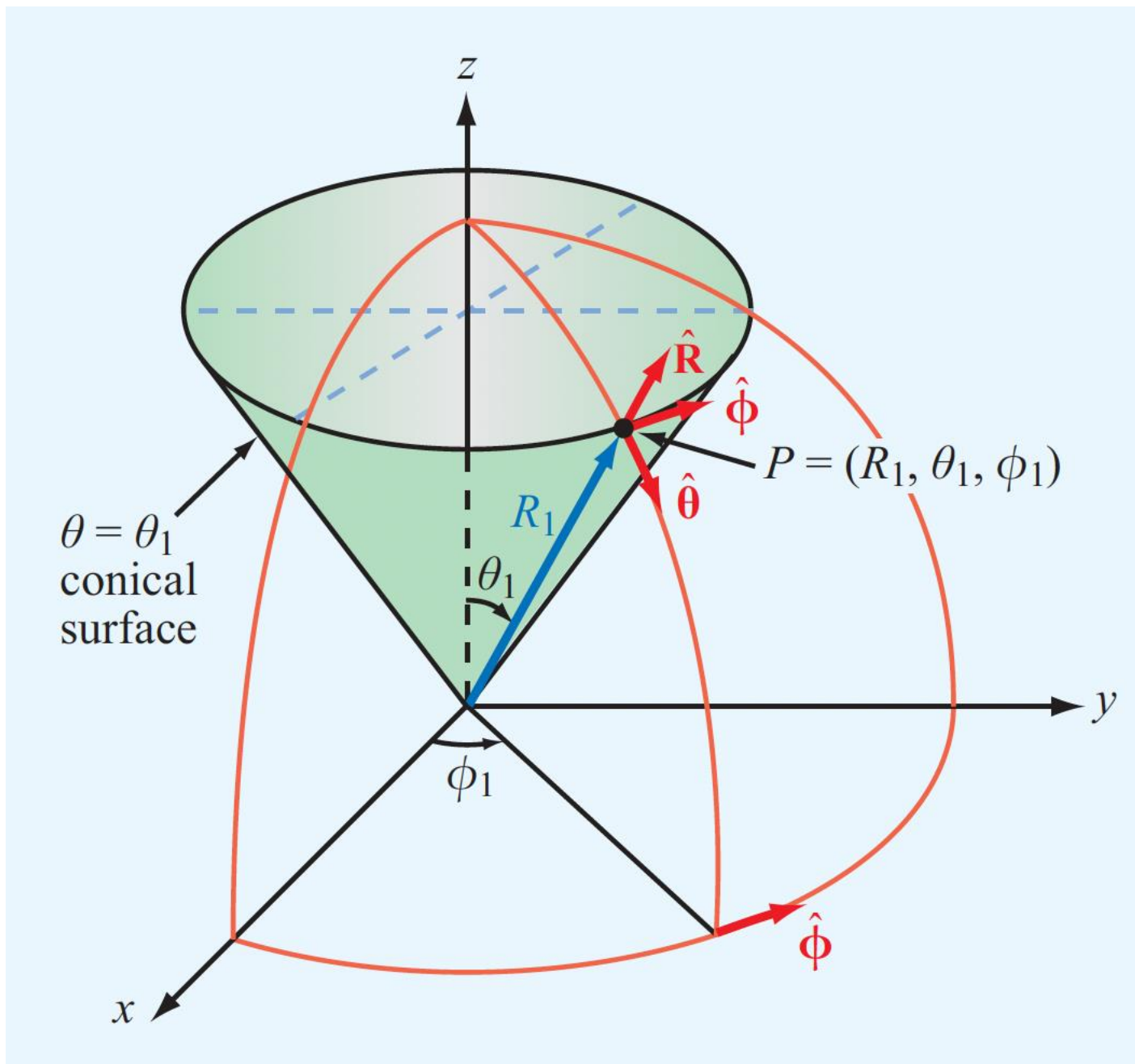


Spherical Coordinate System



Spherical Coordinate System







	Cartesian Coordinates	Cylindrical Coordinates	Spherical Coordinates
Coordinate variables	x, y, z	r, ϕ, z	R, θ, ϕ
Vector representation $\mathbf{A} =$	$\hat{\mathbf{x}}A_x + \hat{\mathbf{y}}A_y + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{r}}A_r + \hat{\boldsymbol{\phi}}A_\phi + \hat{\mathbf{z}}A_z$	$\hat{\mathbf{R}}A_R + \hat{\boldsymbol{\theta}}A_\theta + \hat{\boldsymbol{\phi}}A_\phi$
Magnitude of A $ \mathbf{A} =$	$\sqrt{A_x^2 + A_y^2 + A_z^2}$	$\sqrt{A_r^2 + A_\phi^2 + A_z^2}$	$\sqrt{A_R^2 + A_\theta^2 + A_\phi^2}$
Position vector $\overrightarrow{OP_1} =$	$\hat{\mathbf{x}}x_1 + \hat{\mathbf{y}}y_1 + \hat{\mathbf{z}}z_1,$ for $P(x_1, y_1, z_1)$	$\hat{\mathbf{r}}r_1 + \hat{\mathbf{z}}z_1,$ for $P(r_1, \phi_1, z_1)$	$\hat{\mathbf{R}}R_1,$ for $P(R_1, \theta_1, \phi_1)$
Base vector properties	$\hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{x}} = 0$ $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$ $\hat{\mathbf{y}} \times \hat{\mathbf{z}} = \hat{\mathbf{x}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{x}} = \hat{\mathbf{y}}$	$\hat{\mathbf{r}} \cdot \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1$ $\hat{\mathbf{r}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{r}} = 0$ $\hat{\mathbf{r}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{z}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{z}} = \hat{\mathbf{r}}$ $\hat{\mathbf{z}} \times \hat{\mathbf{r}} = \hat{\boldsymbol{\phi}}$	$\hat{\mathbf{R}} \cdot \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}} \cdot \hat{\boldsymbol{\phi}} = 1$ $\hat{\mathbf{R}} \cdot \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}} \cdot \hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}} \cdot \hat{\mathbf{R}} = 0$ $\hat{\mathbf{R}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\phi}}$ $\hat{\boldsymbol{\theta}} \times \hat{\boldsymbol{\phi}} = \hat{\mathbf{R}}$ $\hat{\boldsymbol{\phi}} \times \hat{\mathbf{R}} = \hat{\boldsymbol{\theta}}$
Dot product $\mathbf{A} \cdot \mathbf{B} =$	$A_x B_x + A_y B_y + A_z B_z$	$A_r B_r + A_\phi B_\phi + A_z B_z$	$A_R B_R + A_\theta B_\theta + A_\phi B_\phi$
Cross product $\mathbf{A} \times \mathbf{B} =$	$\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{r}} & \hat{\boldsymbol{\phi}} & \hat{\mathbf{z}} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$	$\begin{vmatrix} \hat{\mathbf{R}} & \hat{\boldsymbol{\theta}} & \hat{\boldsymbol{\phi}} \\ A_R & A_\theta & A_\phi \\ B_R & B_\theta & B_\phi \end{vmatrix}$
Differential length $d\mathbf{l} =$	$\hat{\mathbf{x}} dx + \hat{\mathbf{y}} dy + \hat{\mathbf{z}} dz$	$\hat{\mathbf{r}} dr + \hat{\boldsymbol{\phi}} r d\phi + \hat{\mathbf{z}} dz$	$\hat{\mathbf{R}} dR + \hat{\boldsymbol{\theta}} R d\theta + \hat{\boldsymbol{\phi}} R \sin \theta d\phi$
Differential surface areas	$ds_x = \hat{\mathbf{x}} dy dz$ $ds_y = \hat{\mathbf{y}} dx dz$ $ds_z = \hat{\mathbf{z}} dx dy$	$ds_r = \hat{\mathbf{r}} r d\phi dz$ $ds_\phi = \hat{\boldsymbol{\phi}} dr dz$ $ds_z = \hat{\mathbf{z}} r dr d\phi$	$ds_R = \hat{\mathbf{R}} R^2 \sin \theta d\theta d\phi$ $ds_\theta = \hat{\boldsymbol{\theta}} R \sin \theta dR d\phi$ $ds_\phi = \hat{\boldsymbol{\phi}} R dR d\theta$
Differential volume $dV =$	$dx dy dz$	$r dr d\phi dz$	$R^2 \sin \theta dR d\theta d\phi$



Transformation	Coordinate Variables	Unit Vectors	Vector Components
Cartesian to cylindrical	$r = \sqrt{x^2 + y^2}$ $\phi = \tan^{-1}(y/x)$ $z = z$	$\hat{\mathbf{r}} = \hat{\mathbf{x}} \cos \phi + \hat{\mathbf{y}} \sin \phi$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_r = A_x \cos \phi + A_y \sin \phi$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$ $A_z = A_z$
Cylindrical to Cartesian	$x = r \cos \phi$ $y = r \sin \phi$ $z = z$	$\hat{\mathbf{x}} = \hat{\mathbf{r}} \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{r}} \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{z}}$	$A_x = A_r \cos \phi - A_\phi \sin \phi$ $A_y = A_r \sin \phi + A_\phi \cos \phi$ $A_z = A_z$
Cartesian to spherical	$R = \sqrt{x^2 + y^2 + z^2}$ $\theta = \tan^{-1}[\sqrt{x^2 + y^2}/z]$ $\phi = \tan^{-1}(y/x)$	$\hat{\mathbf{R}} = \hat{\mathbf{x}} \sin \theta \cos \phi + \hat{\mathbf{y}} \sin \theta \sin \phi + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{x}} \cos \theta \cos \phi + \hat{\mathbf{y}} \cos \theta \sin \phi - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = -\hat{\mathbf{x}} \sin \phi + \hat{\mathbf{y}} \cos \phi$	$A_R = A_x \sin \theta \cos \phi + A_y \sin \theta \sin \phi + A_z \cos \theta$ $A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi - A_z \sin \theta$ $A_\phi = -A_x \sin \phi + A_y \cos \phi$
Spherical to Cartesian	$x = R \sin \theta \cos \phi$ $y = R \sin \theta \sin \phi$ $z = R \cos \theta$	$\hat{\mathbf{x}} = \hat{\mathbf{R}} \sin \theta \cos \phi + \hat{\boldsymbol{\theta}} \cos \theta \cos \phi - \hat{\boldsymbol{\phi}} \sin \phi$ $\hat{\mathbf{y}} = \hat{\mathbf{R}} \sin \theta \sin \phi + \hat{\boldsymbol{\theta}} \cos \theta \sin \phi + \hat{\boldsymbol{\phi}} \cos \phi$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_x = A_R \sin \theta \cos \phi + A_\theta \cos \theta \cos \phi - A_\phi \sin \phi$ $A_y = A_R \sin \theta \sin \phi + A_\theta \cos \theta \sin \phi + A_\phi \cos \phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$
Cylindrical to spherical	$R = \sqrt{r^2 + z^2}$ $\theta = \tan^{-1}(r/z)$ $\phi = \phi$	$\hat{\mathbf{R}} = \hat{\mathbf{r}} \sin \theta + \hat{\mathbf{z}} \cos \theta$ $\hat{\boldsymbol{\theta}} = \hat{\mathbf{r}} \cos \theta - \hat{\mathbf{z}} \sin \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$	$A_R = A_r \sin \theta + A_z \cos \theta$ $A_\theta = A_r \cos \theta - A_z \sin \theta$ $A_\phi = A_\phi$
Spherical to cylindrical	$r = R \sin \theta$ $\phi = \phi$ $z = R \cos \theta$	$\hat{\mathbf{r}} = \hat{\mathbf{R}} \sin \theta + \hat{\boldsymbol{\theta}} \cos \theta$ $\hat{\boldsymbol{\phi}} = \hat{\boldsymbol{\phi}}$ $\hat{\mathbf{z}} = \hat{\mathbf{R}} \cos \theta - \hat{\boldsymbol{\theta}} \sin \theta$	$A_r = A_R \sin \theta + A_\theta \cos \theta$ $A_\phi = A_\phi$ $A_z = A_R \cos \theta - A_\theta \sin \theta$

